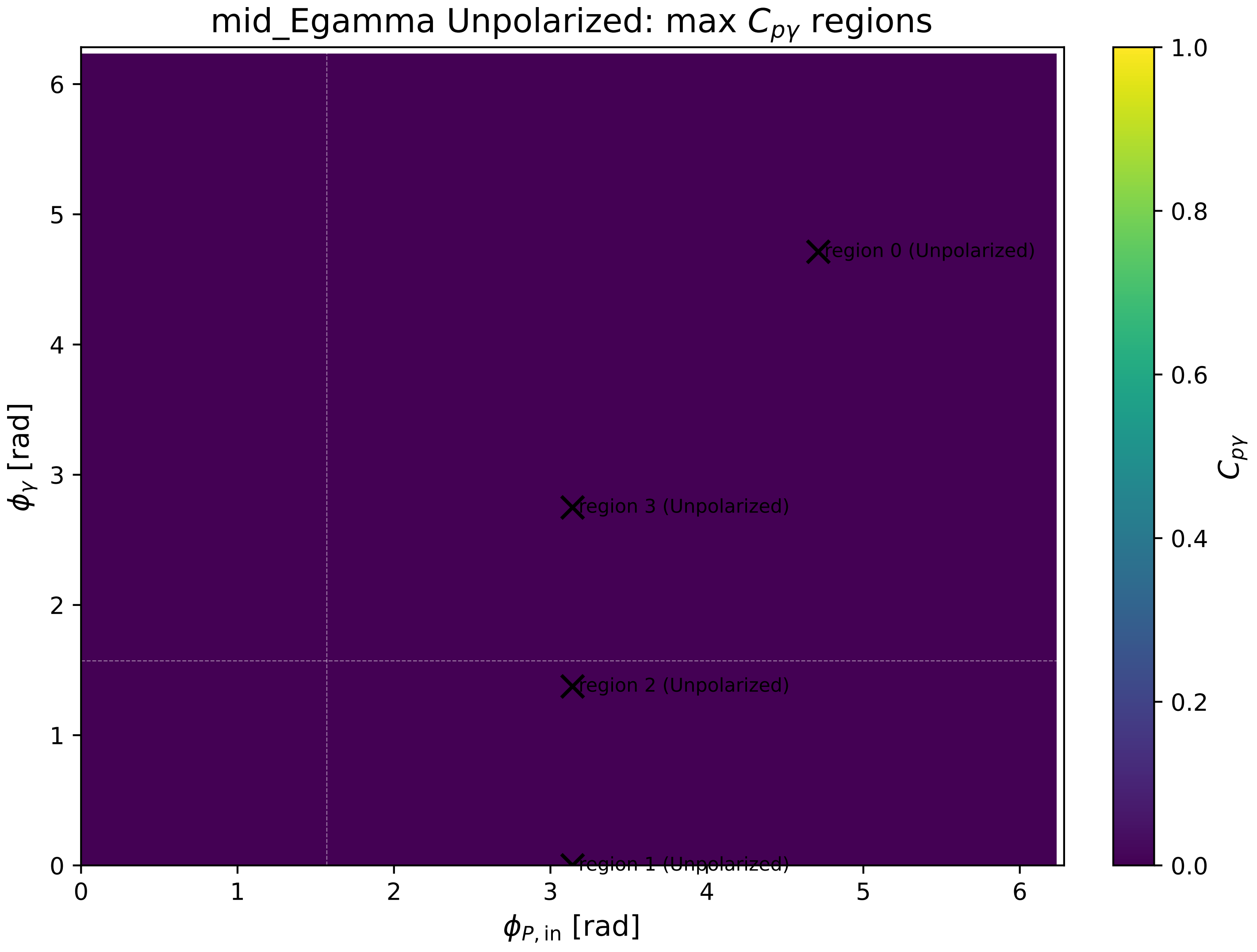
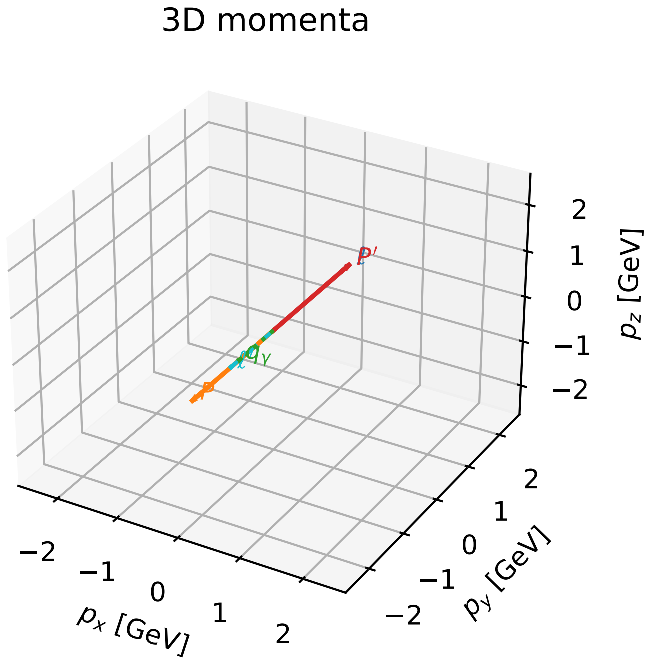


mid_Egamma Unpolarized: max $C_{p\gamma}$ regions
region 0 (Unpolarized)
region 3 (Unpolarized)
region 2 (Unpolarized)
region 1 (Unpolarized)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
$C_{p\gamma}$
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

Unpolarized characteristic max $C_{p\gamma}$ configuration



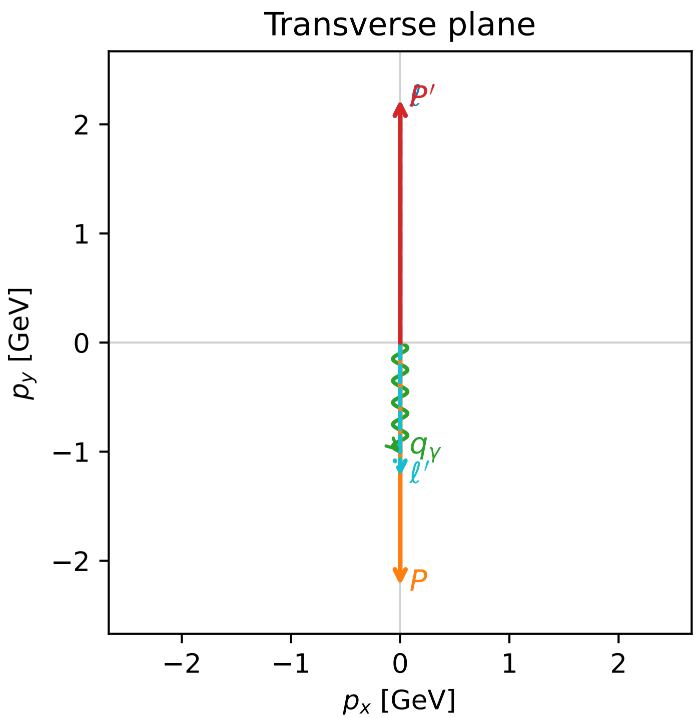

```
c_p_gamma_mid_egamma_unpolarized_region_0  C_{p\gamma}=2.36033e-15
region: mid_Egamma_unpolarized_region_0  final-pair delta_xy=3.14159 rad
outgoing-state purity: 0.5
kinematic point: high_s_high_theta_in_mid_Egamma
```

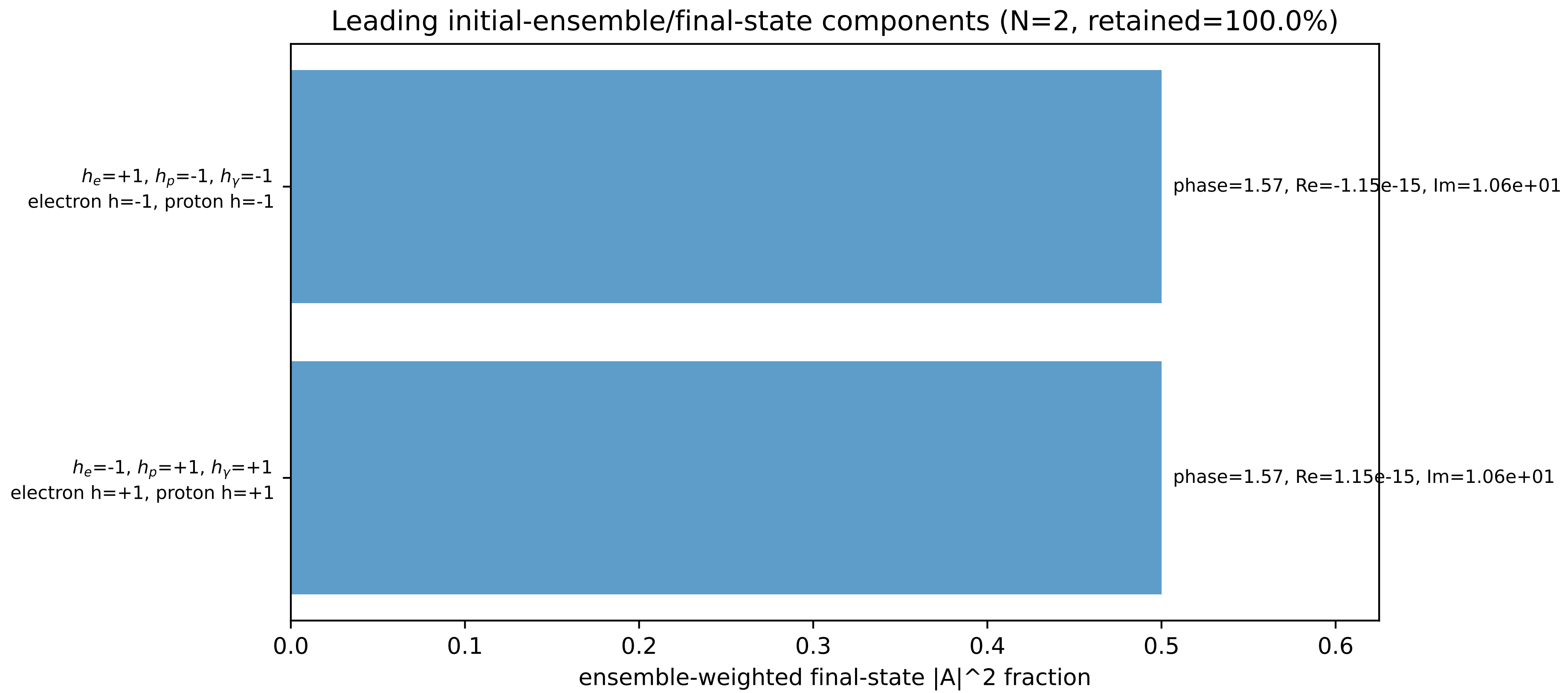
incoming state: $\frac{1}{4}\sum_{h_e,h_p}\rho(h_e,h_p)$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=2.22442
$\theta_{\rm in}$=1.57079, ϕ_ℓ=1.5708, ϕ_P=4.71239
E_γ=1, ϕ_γ=4.71239
Q^2=10.8945, x_B=0.540092, t=-19.7921, W^2=10.1569, y=0.977491
$\theta(\ell',\gamma)$=0 deg

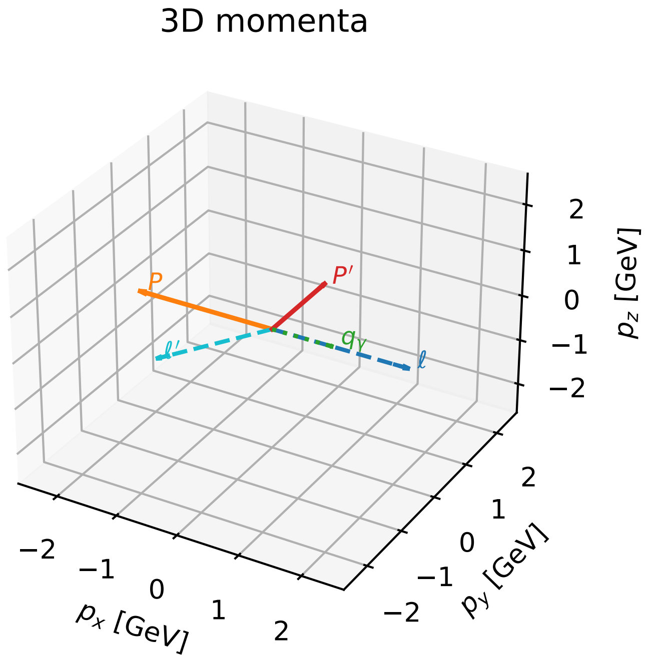
four-momenta [E, p_x, p_y, p_z] GeV:

ℓ	E= 2.2244	p_x= 0.0000	p_y= 2.2244	p_z= -0.0000
P	E= 2.4141	p_x= -0.0000	p_y= -2.2244	p_z= 0.0000
ℓ'	E= 1.2244	p_x= 0.0000	p_y= -1.2244	p_z= 0.0000
P'	E= 2.4141	p_x= 0.0000	p_y= 2.2244	p_z= 0.0000
q_γ	E= 1.0000	p_x= -0.0000	p_y= -1.0000	p_z= 0.0000





Unpolarized characteristic max $C_{p\gamma}$ configuration

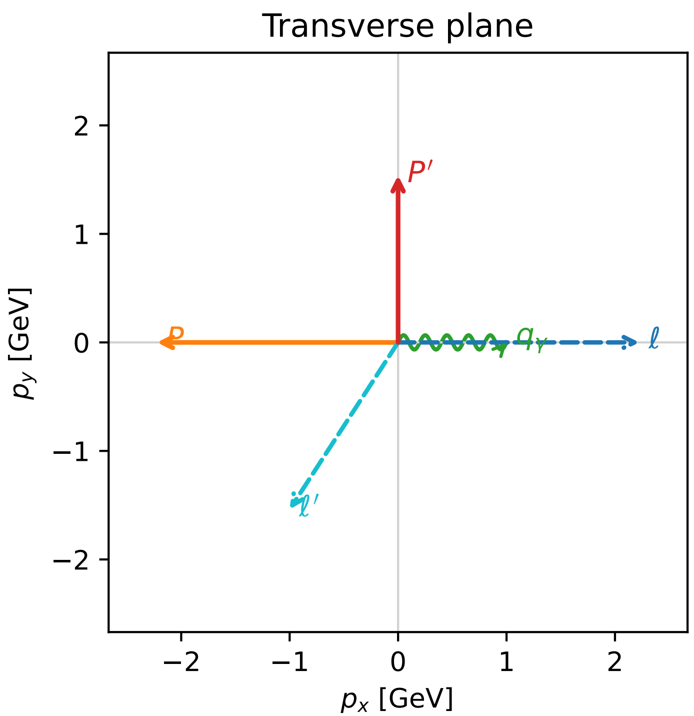


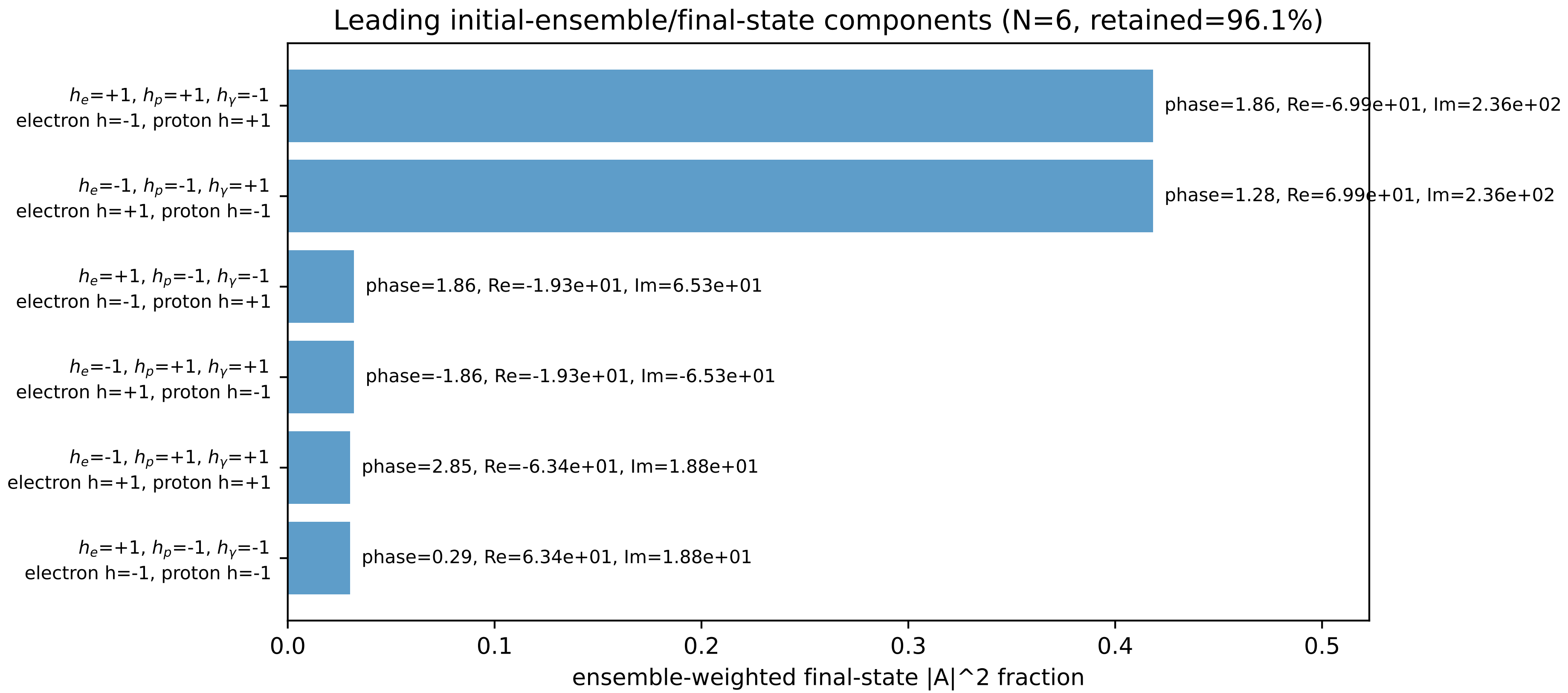

```
c_p_gamma_mid_egamma_unpolarized_region_1  C_{p\gamma}=0
region: mid_Egamma_unpolarized_region_1  final-pair delta_xy=1.5708 rad
outgoing-state purity: 0.424502
kinematic point: high_s_high_theta_in_mid_Egamma
```

incoming state: $\frac{1}{4}\sum_{h_e,h_p}\rho(h_e,h_p)$

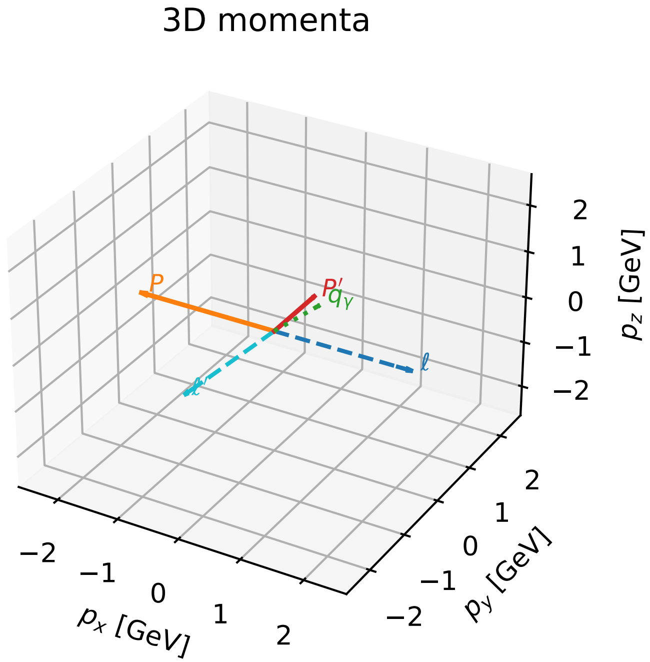
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=1.5395
θ_{in}=1.57079, ϕ_ℓ=0, ϕ_P=3.14159
E_γ=1, ϕ_γ=0
Q^2=12.6159, x_B=0.777732, t=-6.94433, W^2=4.48534, y=0.786071
$\theta(\ell',\gamma)$=123.006 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 2.2244 p_y= -0.0000 p_z= -0.0000
P E= 2.4141 p_x= -2.2244 p_y= 0.0000 p_z= 0.0000
ℓ' E= 1.8358 p_x= -1.0000 p_y= -1.5395 p_z= 0.0000
P' E= 1.8027 p_x= 0.0000 p_y= 1.5395 p_z= 0.0000
q_γ E= 1.0000 p_x= 1.0000 p_y= 0.0000 p_z= 0.0000





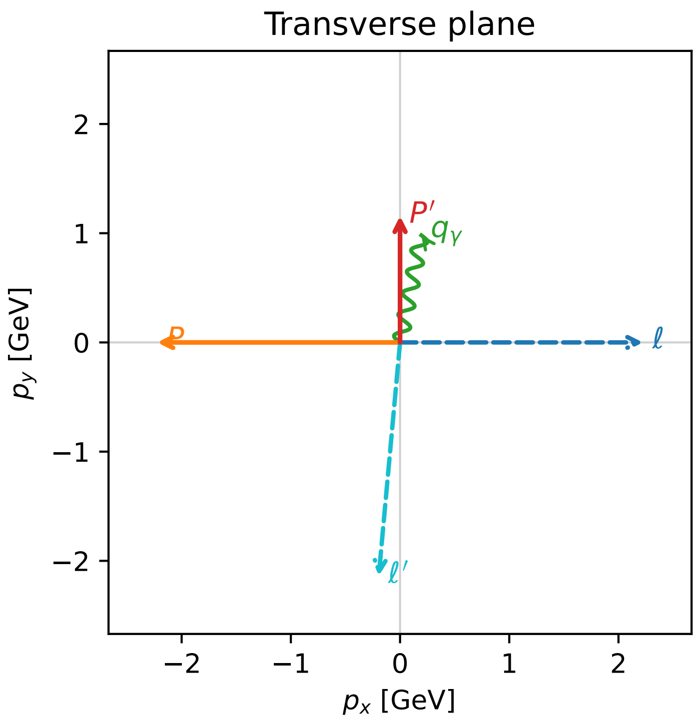
Unpolarized characteristic max $C_{p\gamma}$ configuration

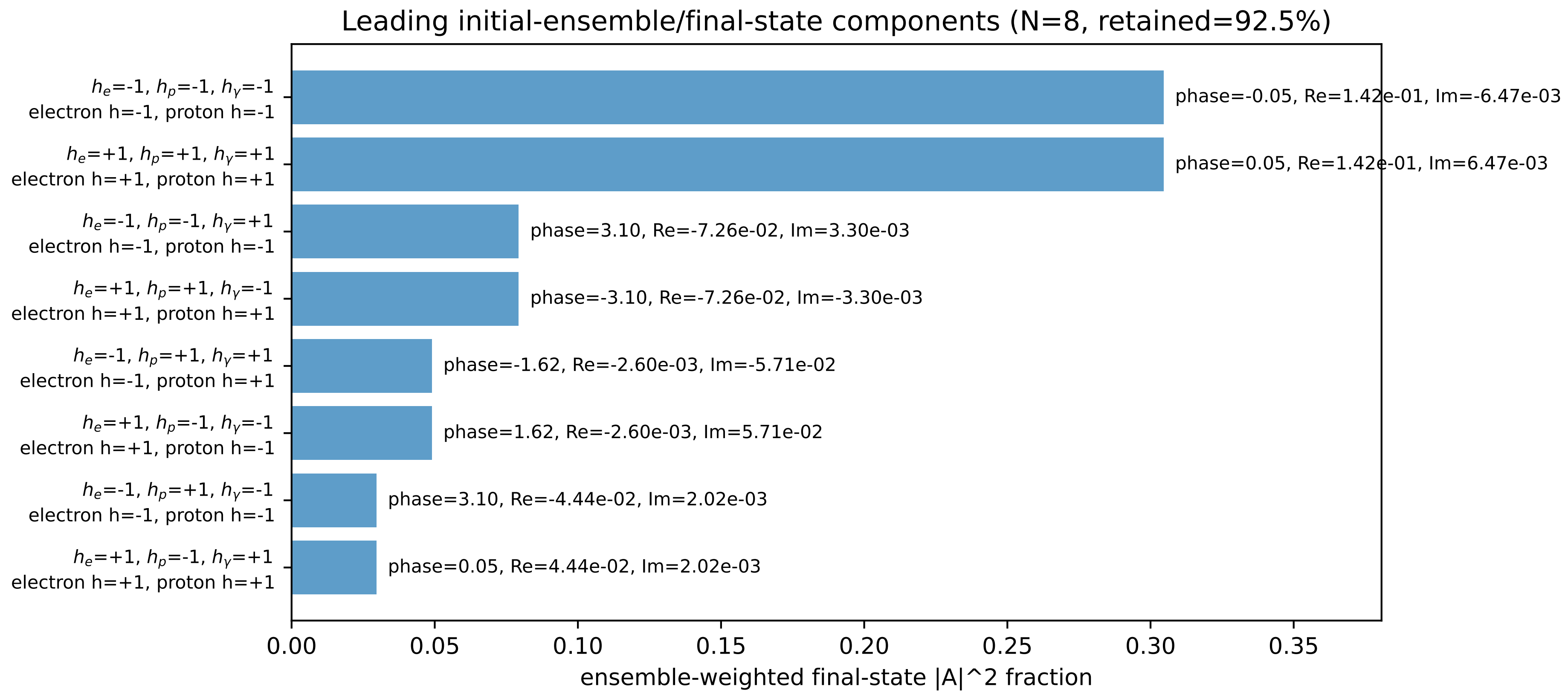


c_p_gamma_mid_egamma_unpolarized_region_2 C_{p\gamma}=0
region: mid_Egamma_unpolarized_region_2 final-pair delta_xy=0.19635 rad
outgoing-state purity: 0.37737
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $\frac{1}{4}\sum_{h_e,h_p}\rho(h_e,h_p)$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=1.15835
$\theta_{\rm in}$=1.57079, ϕ_ℓ=0, ϕ_P=3.14159
E_γ=1, ϕ_γ=1.37445
Q^2=10.4241, x_B=0.93633, t=-5.43678, W^2=1.58868, y=0.539489
$\theta(\ell',\gamma)$=173.961 deg

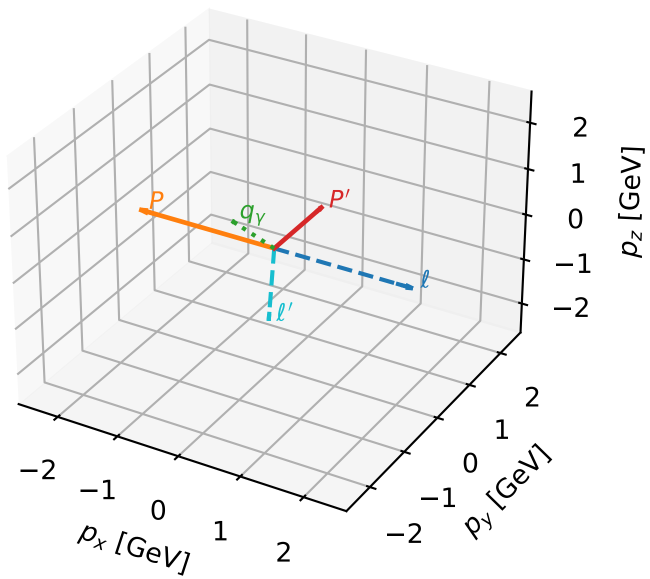
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 2.2244 p_y= -0.0000 p_z= -0.0000
P E= 2.4141 p_x= -2.2244 p_y= 0.0000 p_z= 0.0000
ℓ' E= 2.1480 p_x= -0.1951 p_y= -2.1391 p_z= 0.0000
P' E= 1.4905 p_x= 0.0000 p_y= 1.1583 p_z= 0.0000
q_γ E= 1.0000 p_x= 0.1951 p_y= 0.9808 p_z= 0.0000





Unpolarized characteristic max $C_{p\gamma}$ configuration

3D momenta

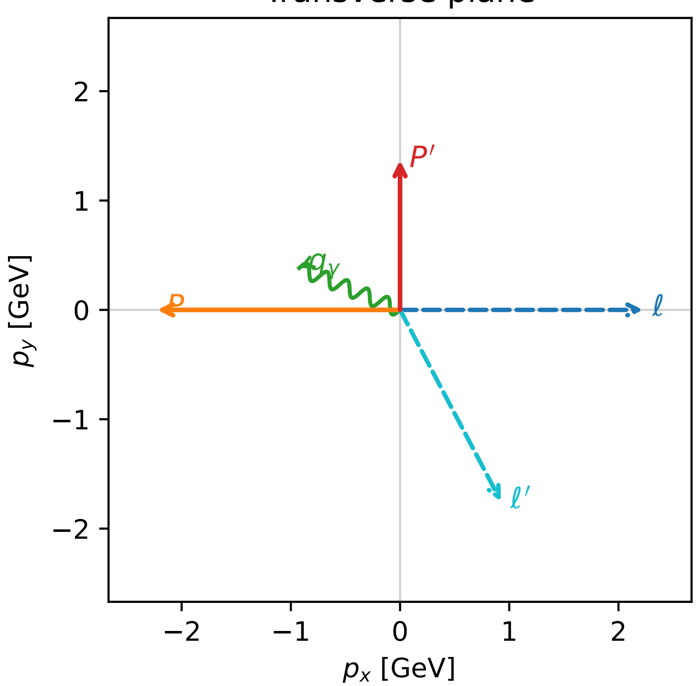


c_p_gamma_mid_egamma_unpolarized_region_3 C_{p\gamma}=0
region: mid_Egamma_unpolarized_region_3 final-pair delta_xy=1.1781 rad
outgoing-state purity: 0.303247
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $\frac{1}{4}\sum_{h_e,h_p}\rho(h_e,h_p)$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=1.36819
θ_{in}=1.57079, ϕ_ℓ=0, ϕ_P=3.14159
E_γ=1, ϕ_γ=2.74889
Q^2=4.69704, x_B=0.674129, t=-6.24956, W^2=3.15038, y=0.337641
$\theta(\ell',\gamma)$=140.319 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 2.2244 p_y= -0.0000 p_z= -0.0000
P E= 2.4141 p_x= -2.2244 p_y= 0.0000 p_z= 0.0000
ℓ' E= 1.9797 p_x= 0.9239 p_y= -1.7509 p_z= 0.0000
P' E= 1.6588 p_x= 0.0000 p_y= 1.3682 p_z= 0.0000
q_γ E= 1.0000 p_x= -0.9239 p_y= 0.3827 p_z= 0.0000

Transverse plane



Leading initial-ensemble/final-state components (N=8, retained=90.5%)